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AI/ML Intern

Final Research Documentation

1. Complete AI Workflow

The proposed AI-powered virtual try-on workflow consists of the following stages:

User Image Upload



Image Preprocessing

Resize image

Improve image quality

Background preparation



Human Segmentation

Separate user from background

Generate person mask



Pose Detection

Detect body landmarks

Identify body orientation



Body Landmark Extraction

Shoulder points

Waist points

Hip points

Arm positions



Body Measurement Estimation

Shoulder width

Height estimation

Waist positioning

Body proportions



Garment Processing

Load garment image/model

Extract garment features

Prepare garment alignment



Virtual Try-On Generation

Align garment with body

Generate try-on visualization

Produce final output image



Recommendation Engine

Size recommendation

Product recommendation

Outfit suggestions



Final Output

Model Comparison Report

Model	Purpose	Advantages	Limitations
MediaPipe Pose	Pose Detection & Body Landmarks	Fast, lightweight, real-time	Less accurate for extreme poses
MediaPipe Selfie Segmentation	Human Segmentation	Segmentation Easy implementation, good performance	Sensitive to complex backgrounds
CP-VTON	2D Virtual Try-On	Faster processing, lower resource usage	Limited realism
VITON-HD	High-Resolution Virtual Try-On	Better garment fitting and visual quality	Higher computational requirements
Trimesh + Pyrender	3D Model Visualization	Useful for avatar and garment rendering	Not a complete virtual try-on solution
Unity	3D Virtual Try-On Platform	AR/VR support, real-time rendering	Requires additional development effort

Key Observation

For lightweight virtual try-on systems, MediaPipe-based workflows are effective. For advanced virtual fitting experiences, VITON-HD and Unity-based 3D implementations offer greater realism but require more computational resources.

Future Improvement Suggestions for RARITONE

Based on the research conducted during the internship, the following improvements could enhance future virtual try-on solutions:

1. Virtual Fashion Analytics Dashboard

Provide brands with anonymous insights about:

- Most tried-on products
- Popular sizes
- Fit issues
- Customer preferences

Benefit: Valuable business intelligence for retailers.

2. Digital Twin Wardrobe

Create a persistent digital avatar for each user that stores measurements, preferences, and previous try-ons.

Benefit: Faster future shopping experiences and personalized recommendations.

3. AI Fit Confidence Score

Develop an AI score that indicates how confidently a garment will fit a user based on body measurements, garment dimensions, and historical fitting data.

Benefit: Reduces return rates and increases customer trust.

Project Summary

The internship focused on researching technologies required for AI-powered virtual clothing try-on systems. The work involved studying human segmentation, pose estimation, body landmark detection, body measurement extraction, virtual try-on architectures, datasets, and recommendation systems.

Various models and frameworks were evaluated, including MediaPipe, CP-VTON, VITON-HD, Trimesh, Pyrender, Unity, and AR-based platforms. Public datasets such as DeepFashion, DeepFashion2, VITON, Dress Code, and Fashionpedia were also analyzed.

The research helped establish a complete workflow showing how computer vision and machine learning techniques can be combined to create realistic virtual fitting experiences for online fashion applications.

Conclusion

The internship provided valuable exposure to Artificial Intelligence, Machine Learning, Computer Vision, and Virtual Try-On technologies. Through model evaluation, workflow design, dataset analysis, and technology research, a strong understanding of modern virtual try-on systems was developed.

The findings demonstrate that combining segmentation, pose estimation, body analysis, and recommendation systems can significantly improve online shopping experiences. The knowledge gained during this internship provides a solid foundation for future work in AI-driven fashion technology and intelligent recommendation systems.